National Textile University

Department of Computer Science

Subject:
Data Science

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Reg number:
23-NTU-CS-1201

Assignment no.:
02

Semester:
6th

Data Science

Assignment 02

Measuring Data Similarity and Dissimilarity

Data Set

Object	Nominal	Ordinal	Numeric	Binary
1	Code A	Excellent	45	Yes
2	Code B	Fair	22	No
3	Code C	Good	64	Yes
4	Code D	Excellent	28	No

1. Nominal Attribute

For nominal attribute we simply check whether the values match or not. If two objects have the same value

If distance is equal to zero and If values are different than distance is equal to one.

	1	2	3	4
1	0			
2	1	0		
3	1	1	0	
4	0	1	1	0

2. Ordinal Attribute

Step 1: Convert Categories into ranks

Fair = 1, good = 2, excellent = 3

Step 2: Normalize this ranks using:

$$z = \frac{r-1}{M-1}$$ where M = number of states = 3

Object	Rank	Normalized value
1	3	1.0
2	1	0.0
3	2	0.5
4	3	1.0

3. Numeric Attribute

Numeric dissimilarity is calculated using normalization:

$$d(i,j) = \frac{|x_i - x_j|}{(max - min)\, max} = 22$$

range = 42

	1	2	3	4
1	0			
2	0.55	0		
3	0.45	1.00	0	
4	0.40	0.14	0.86	0

4. Binary Attibute

Binary Attributes contain only two states Yes or No. We convert Yes = 1 and No = 0.

Object values become:

$1 \rightarrow 1,\ 2 \rightarrow 0,\ 3 \rightarrow 1,\ 4 \rightarrow 0$

	1	2	3	4
1	0			
2	1	0		
3	0	1	0	
4	1	0	1	0

5. Final Mixed Attirbute

To compute final distance we combine all attributes.

Formula:

$$d(i,j) = (\text{Numeric} + \text{Nominal} + \text{Ordinal} + \text{Binary})/4$$

Distance between Object 1 and Object 2

$= (1+1+0.55+1)/4$

$= 0.8875 \approx 0.89$

	1	2	3	4
1	0			
2	0.89	0		
3	0.49	0.88	0	
4	0.35	0.54	0.84	0

Conclusion:

From final dissimilarity matrix we observe that Object 1 and Object 4 have the smallest distance which means they are the most similar records. Object 1 and Object 2 have the highest distance which means they are the most dissimilar. This example clearly demonstrates how mixed type of attributes can be combined to measure similarity in data mining.